Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 10.01

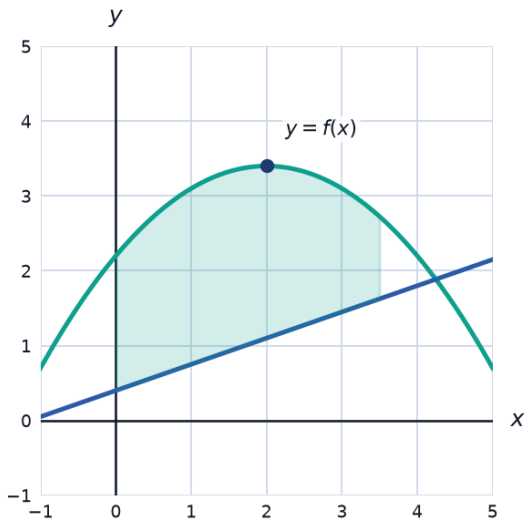
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Review: Circumference and area

In the diagram below, two chords, AB and CD, intersect at point P inside a circle. If $AP = 5$ cm, $PB = 8$ cm, and $CP = 4$ cm, what is the length of PD in cm?

2 [Foundational] • Target: Arc length and sector area

In the figure below with $AB \parallel CD$, $\text{Area}(\Delta ABE) = 36$ and $\text{Area}(\Delta CDE) = 81$. If base $AB = 10$, what is the length of base CD?



3 [Foundational] • Target: Extension: Chords of a circle

The triangle ABC is translated on the coordinate plane according to the rule $(x, y) \to (x - 3, y + 2)$. If the vertices of the image triangle A'B'C' are A'(-1, 5), B'(3, 1), and C'(-2, -3), what are the coordinates of the original vertices A, B, and C?

4 [Foundational] • Target: Inscribed angles and polygons

In ΔABC, $AD = 3$ and $DB = 9$ (so $AB = 12$); $AE = 4$ and $EC = 5$ (so $AC = 9$). If base $BC = 21$, what is the length of segment DE?

5 [Foundational] • Target: Angles from intersecting chords, secants, and tangents

The diagonals of trapezoid $ABCD$ with $AB \parallel CD$ intersect at P. If $\text{Area}(\Delta APB) = 16$ and $\text{Area}(\Delta CPD) = 49$, what is the total area of trapezoid $ABCD$?

6 [Foundational] • Target: Tangents to a circle

In ΔABC, segment $DE \parallel BC$ with D on AB and E on AC. If $AD = 5$, $DB = 10$, and $DE = 7$, what is the length of base BC?

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7 [Foundational] • Target: Review: Circumference and area

In the hourglass figure below, segment $AB \parallel CD$, and segments AC and BD intersect at E. If $AB = 8$, $CD = 12$, and $AE = 6$, what is the length of segment CE?

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Arc length and sector area

A right triangle has legs of lengths 15 and 20. What is the exact length of the altitude drawn perpendicular to the hypotenuse?

9 [Intermediate] • Target: Extension: Chords of a circle

In ΔABC, $\angle ADE \cong \angle C$ with $AD = 5$ and $AC = 15$. If the total area of ΔABC is 108, what is the area of quadrilateral $BDEC$?

10 [Intermediate] • Target: Inscribed angles and polygons

In ΔABC shown below, segment $DE \parallel BC$. If $AD = 6$, $DB = 9$, and $AE = 8$, what is the length of segment EC?

11 [Intermediate] • Target: Angles from intersecting chords, secants, and tangents

In ΔABC, $DE \parallel BC$ with D on AB and E on AC. If $AD = 4$ and $AB = 10$, what fraction of the area of ΔABC is occupied by trapezoid $BDEC$?

12 [Intermediate] • Target: Tangents to a circle

Trapezoid $ABCD$ has parallel bases $AB = 6$ and $CD = 14$. The diagonals AC and BD intersect at point P. What is the ratio of $\text{Area}(\Delta APB)$ to $\text{Area}(\Delta CPD)$?

13 [Intermediate] • Target: Review: Circumference and area

In ΔABC, point D lies on AB and E on AC such that $\angle AED \cong \angle B$. If $AE = 6$, $AD = 8$, $AC = 18$, and base $BC = 27$, what is the length of segment DE?

14 [Intermediate] • Target: Arc length and sector area

In trapezoid $ABCD$ with $AB \parallel CD$, $AB = 8$ and $CD = 12$. The diagonals intersect at P. If the entire diagonal $AC = 30$, what is the length of segment AP?

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Extension: Chords of a circle • Cross-Lesson: 5.01 Right Triangles & Trigonometry

In the bowtie figure with $AB \parallel CD$, segments AC and BD intersect at E. If $AB = 10$, $CD = 15$, and the entire segment $BD = 35$, what is the length of segment BE?

16 [Challenging] • Target: Inscribed angles and polygons • Cross-Lesson: 6.01 Polygons & Quadrilaterals

In ΔABC, ray AD bisects $\angle A$ with point D on side BC. If $AB = 8$, $AC = 12$, and $BD = 6$, what is the length of segment DC?

17 [Challenging] • Target: Angles from intersecting chords, secants, and tangents • Cross-Lesson: 7.01 Circles

In ΔABC, ray AD bisects $\angle A$. If $AB = x$, $AC = x + 4$, $BD = 4$, and $DC = 6$, what is the length of side AC?

18 [Challenging] • Target: Tangents to a circle • Cross-Lesson: 8.01 Area, Surface Area & Volume

In right ΔABC, altitude CD to hypotenuse AB measures 6, and the entire hypotenuse $AB = 13$. What is the length of the longer of the two segments into which D divides AB?

19 [Challenging] • Target: Review: Circumference and area • Cross-Lesson: 9.01 Transformations & Symmetry

In ΔABC, segment $DE \parallel BC$. If $AD = 8$, $DB = 4$, and $AC = 18$, what is the length of segment AE?

20 [Challenging] • Target: Arc length and sector area • Cross-Lesson: 1.01 Foundations & Reasoning

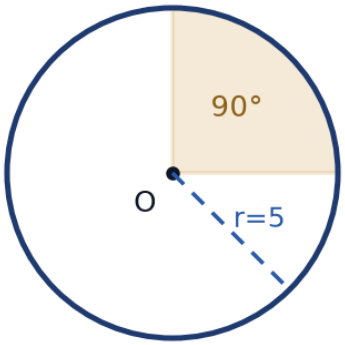
In ΔABC, ray AD bisects $\angle A$ with D on BC. If $AB = 10$, $AC = 15$, and the entire base $BC = 20$, what is the length of segment BD?

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Review: Circumference and area

For the model $f(x) = 6x + 6$ in review: circumference and area, evaluate $f(2)$.



a. 18

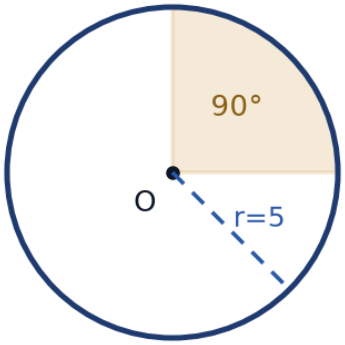
b. 24

c. 12

d. 22

22 [Foundational] • Target: Review: Circumference and area

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



a. $x = 2$ and $x = 8$

b. $x = 2$ and $x = 7$

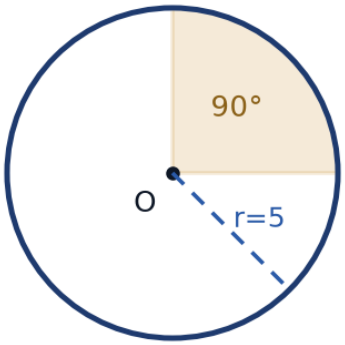
c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

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23 [Foundational] • Target: Review: Circumference and area

For the model $f(x) = 6x + 6$ in review: circumference and area, evaluate $f(5)$.



a. 36

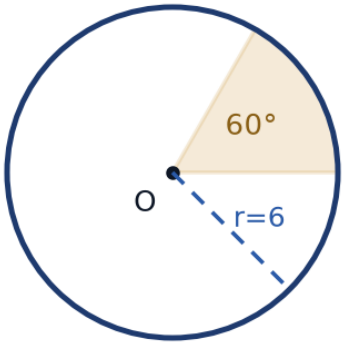
b. 42

c. 30

d. 40

24 [Foundational] • Target: Arc length and sector area

For the model $f(x) = 2x + 7$ in arc length and sector area, evaluate $f(2)$.



a. 11

b. 13

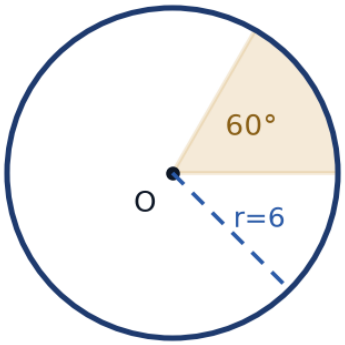
c. 4

d. 15

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25 [Foundational] • Target: Arc length and sector area

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

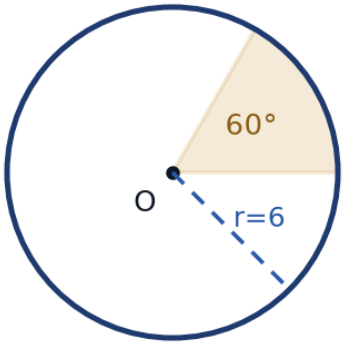
b. $x = 2$ and $x = 4$

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

26 [Foundational] • Target: Arc length and sector area

For the model $f(x) = 2x + 7$ in arc length and sector area, evaluate $f(5)$.



a. 17

b. 19

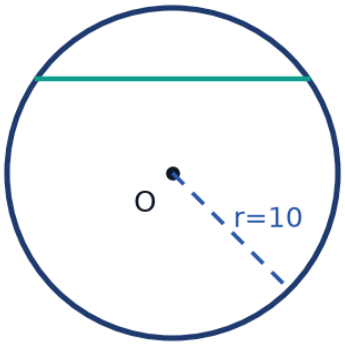
c. 10

d. 21

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27 [Foundational] • Target: Extension: Chords of a circle

For the model $f(x) = 3x + 8$ in extension: chords of a circle, evaluate $f(2)$.



a. 14

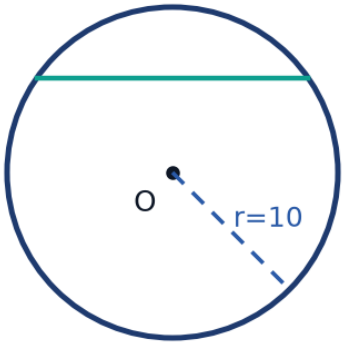
b. 17

c. 6

d. 18

28 [Foundational] • Target: Extension: Chords of a circle

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Review: Circumference and area**

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ and $x = 6$

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

30 **[Intermediate]** • **Target: Review: Circumference and area**

For the model $f(x) = 6x + 6$ in review: circumference and area, evaluate $f(3)$.

a. 24

b. 30

c. 18

d. 28

31 **[Intermediate]** • **Target: Review: Circumference and area**

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ and $x = 8$

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

32 **[Intermediate]** • **Target: Arc length and sector area**

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

33 **[Intermediate]** • **Target: Arc length and sector area**

For the model $f(x) = 2x + 7$ in arc length and sector area, evaluate $f(3)$.

a. 13

b. 15

c. 6

d. 17

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34 [Intermediate] • Target: Arc length and sector area

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

35 [Intermediate] • Target: Extension: Chords of a circle

A data set recorded for extension: chords of a circle consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5

d. 25.5

36 [Intermediate] • Target: Extension: Chords of a circle

For the model $f(x) = 3x + 8$ in extension: chords of a circle, evaluate $f(4)$.

a. 20

b. 23

c. 12

d. 24

37 [Intermediate] • Target: Extension: Chords of a circle

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Inscribed angles and polygons • Cross-Lesson: 8.01 Area, Surface Area & Volume

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 15

c. 13

d. 17

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39 [Challenging] • Target: Inscribed angles and polygons • Cross-Lesson: 9.01 Transformations & Symmetry

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 9)$. What is its y-coordinate when $x = 2$?

a. 13

b. 19

c. 17

d. 21

40 [Challenging] • Target: Inscribed angles and polygons • Cross-Lesson: 1.01 Foundations & Reasoning

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm

c. 30 cm

d. 96 cm

41 [Challenging] • Target: Angles from intersecting chords, secants, and tangents • Cross-Lesson: 11.01 Probability

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

42 [Challenging] • Target: Angles from intersecting chords, secants, and tangents • Cross-Lesson: 2.01 Lines & Angles

In ΔABC studying angles from intersecting chords, secants, and tangents, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 85°

b. 95°

c. 77°

d. 137°

43 [Challenging] • Target: Angles from intersecting chords, secants, and tangents • Cross-Lesson: 3.01 Triangles & Congruence

In ΔABC studying angles from intersecting chords, secants, and tangents, $m\angle A = 43^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 78°

b. 88°

c. 70°

d. 137°

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44 [Challenging] • Target: Angles from intersecting chords, secants, and tangents • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm

c. 42 cm

d. 175 cm

45 [Challenging] • Target: Tangents to a circle • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A data set recorded for tangents to a circle consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

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